

BPSC TRE-4.0 2026

Higher Secondary Teacher – Classes 11–12

DETAILED TOPIC-WISE SYLLABUS & PREPARATION BREAKDOWN

Higher-secondary paper: 30 + 40 + 80 structure. The subject list is the widest; it includes languages, Mathematics, Physics, Chemistry, Botany, Zoology, History, Geography, Political Science, Economics, Sociology, Psychology, Philosophy, Home Science, Computer Science, Business Studies, Accountancy, Music, Fine Arts, Dance, Physical Education and other listed languages/specialist subjects.

Exam Pattern / Paper Structure

Part	Component	Questions / Marks
Part-I	Language (Qualifying)	30 / 30
Part-II	General Studies	40 / 40
Part-III	Concerned Subject	80 / 80
Total	—	150 / 150

Part-I / Language

English

- Reading comprehension: factual, literary and discursive passages; main idea, inference, tone, vocabulary and contextual meaning.
- Parts of speech: nouns, pronouns, adjectives, verbs, adverbs, prepositions, conjunctions and interjections.
- Tenses, subject-verb agreement, articles, determiners, modals, auxiliaries and question forms.
- Active/passive voice, direct/indirect speech, sentence transformation, clauses and degrees of comparison.
- Vocabulary: synonyms, antonyms, homonyms, one-word substitutions, idioms and phrases.
- Sentence correction, error detection, fill in the blanks, rearrangement, cloze tests and spelling/usage.
- Basic functional writing and sentence formation.

Hindi / Urdu / Bengali (applicable language option)

- Reading comprehension and interpretation of prose and poetry.
- Grammar: nouns, pronouns, adjectives, verbs, adverbs, conjunctions, prepositions and sentence structure.
- Gender, number, case, tense, voice, sentence correction and common usage.
- Sandhi, Samasa, prefixes/suffixes or their language-specific equivalents where applicable.
- Vocabulary: synonyms, antonyms, homonyms, idioms, proverbs and contextual meanings.
- Spelling, error correction, sentence transformation, fill-in-the-blanks and language usage.
- Basic literature awareness and prescribed school-level prose/poetry where applicable.

Part-II / General Studies

Elementary Mathematics

- Number system: natural numbers, whole numbers, integers, rational numbers, fractions, decimals, place value, divisibility rules, factors and multiples, HCF and LCM.
- Simplification and approximation: BODMAS, fractions/decimals, squares, cubes, square roots and estimation.
- Ratio and proportion: equivalent ratios, direct proportion, inverse proportion, unitary method and applications.
- Percentage: conversion between fractions/decimals/percentages, percentage increase/decrease, successive changes and applications.
- Profit, loss and discount: cost price, selling price, marked price, discount, profit/loss percentage and successive discounts.
- Interest: simple interest, compound interest, amount, principal, rate, time and basic applications.
- Average: arithmetic mean, weighted/basic averages and age-based applications.
- Time and work: work efficiency, combined work, wages, pipes and cisterns.
- Time, speed and distance: basic speed, average speed, relative speed, trains, boats and streams.
- Algebra: algebraic expressions, identities, linear equations, simple simultaneous equations, powers and exponents.
- Geometry: lines, angles, triangles, congruence, similarity, quadrilaterals, polygons, circles and basic constructions.
- Mensuration: perimeter, area, surface area and volume of common 2D/3D figures.
- Data handling: tables, pictographs, bar graphs, line graphs, pie charts, mean, median, mode and basic probability.

Mental Ability Test

- Analogy: word, number, alphabet and figure analogies.
- Classification: odd-one-out based on words, numbers, letters and figures.
- Series: number, alphabet, mixed and pattern series; missing-term questions.
- Coding-decoding: letter coding, number coding, substitution and pattern coding.
- Blood relations: family trees, coded relations and generation-based questions.
- Direction sense: compass directions, turns, distance and route tracing.
- Ranking and ordering: position, age, height, marks and comparison problems.
- Syllogism: statements, conclusions, logical deduction and basic Venn-diagram representation.
- Venn diagrams, calendars, clocks, missing numbers, simple puzzles and non-verbal reasoning.

General Awareness

- India and neighbouring countries: geography, history, capitals, currencies, important institutions and major bilateral/regional developments.
- Current affairs: national and international events, government schemes, appointments, awards, books and authors, reports, summits, important days and recent developments.
- Indian history and culture: ancient, medieval and modern periods; art, literature, heritage, festivals and important personalities.
- Indian polity: Constitution, Preamble, Fundamental Rights, Fundamental Duties, Directive Principles, Parliament, President, Prime Minister, judiciary, elections, federalism and local government.
- Indian economy: basic concepts, sectors, banking, inflation, taxation, budget, development indicators, poverty, unemployment and major welfare schemes.
- Sports: major tournaments, trophies, players, records and sports-related current affairs.
- Environment and ecology: biodiversity, conservation, pollution, climate change, sustainable development and environmental institutions.

General Science

- Physics: measurement, units, motion, force, work, energy, power, gravitation, pressure, heat, temperature, sound, light, reflection, refraction, electricity and magnetism.
- Chemistry: matter, physical/chemical changes, atoms, molecules, atomic structure, elements, compounds, mixtures, periodic classification, chemical reactions, acids, bases, salts, metals, non-metals and carbon compounds.
- Biology: cell, tissues, plants, animals, nutrition, respiration, transportation, reproduction, heredity basics, microorganisms, human body systems, health, diseases and immunity.

- Environment: ecosystem, food chain, food web, natural resources, pollution, conservation, biodiversity and climate change.

Social Science

- History: ancient India, medieval India, colonial India, major dynasties, social and cultural developments and the freedom movement.
- Civics: democracy, institutions, elections, rights, Constitution, federalism, local government and citizen participation.
- Economics: sectors of the economy, poverty, unemployment, money and credit, development, public facilities, markets and sustainable development.
- Social issues: diversity, inequality, gender, education, health, livelihoods, population and social change.

Geography and Environment

- Solar system: Sun, planets, satellites, Earth and basic astronomical concepts.
- Earth: shape, motions, latitudes, longitudes, time zones, maps, scale and directions.
- Physical geography: continents, mountains, plateaus, plains, deserts, rivers, lakes, oceans, volcanoes and earthquakes.
- Indian geography: physiographic divisions, Himalayas, plains, plateau, coastal areas, islands, rivers, climate, soils, vegetation and natural resources.
- Human geography: population, settlements, migration, occupations, transport, communication and human activities.
- Environment: ecosystems, natural resources, pollution, waste management, conservation, biodiversity, climate change and disaster awareness.

Indian National Movement

- British expansion and consolidation in India; major administrative and economic changes.
- Revolt of 1857: causes, major centres, leaders, nature and consequences.
- Socio-religious reform movements, modern education, press and growth of political awareness.
- Rise of Indian nationalism and formation of the Indian National Congress.
- Partition of Bengal, Swadeshi and Boycott, revolutionary activities and Home Rule movement.
- Gandhian era: Champaran, Kheda, Ahmedabad, Non-Cooperation, Civil Disobedience and Quit India movements.
- Role of major leaders, women, peasants, workers and tribal communities in the freedom struggle.
- Constitutional developments, Cabinet Mission, transfer of power and independence.



Part-III / Concerned Subject

Select the subject applicable to the post. The following is a detailed preparation breakdown. The official notification identifies the subject areas and curriculum level; the chapter lists below expand those areas into study-ready topics and should not be treated as verbatim notification wording.

Hindi

- Comprehension of prose and poetry; main idea, inference, vocabulary and interpretation.
- Grammar: sandhi, samasa, prefixes, suffixes, noun, pronoun, adjective, verb, case, tense, voice and sentence structure.
- Synonyms, antonyms, one-word substitutions, idioms, proverbs, spelling and sentence correction.
- Figures of speech, rasa, metre, poetic devices and basic literary terminology.
- School-level prose, poetry, story, essay, drama and major literary forms according to the prescribed curriculum.
- Literary history, major literary movements, important authors/poets, genres and prescribed texts.
- Advanced grammar, poetic devices, rasa, chhand, alankar, sentence transformation and language analysis.

English

- Reading comprehension: factual, literary and discursive passages; main idea, inference, tone, vocabulary and contextual meaning.
- Parts of speech: nouns, pronouns, adjectives, verbs, adverbs, prepositions, conjunctions and interjections.
- Tenses, subject-verb agreement, articles, determiners, modals, auxiliaries and question forms.
- Active/passive voice, direct/indirect speech, sentence transformation, clauses and degrees of comparison.
- Vocabulary: synonyms, antonyms, homonyms, one-word substitutions, idioms and phrases.
- Sentence correction, error detection, fill in the blanks, rearrangement, cloze tests and spelling/usage.
- Basic functional writing and sentence formation.
- Literature: poetry, prose, drama, fiction, theme, plot, character, setting, tone, symbolism and literary devices.
- Literary periods and major writers/works as prescribed; comprehension and critical interpretation.
- Writing: essay/paragraph, letter/application, précis/summary, editing, rearrangement and functional English.

Urdu

- Reading comprehension and interpretation of prose and poetry.
- Grammar: nouns, pronouns, adjectives, verbs, adverbs, conjunctions, prepositions and sentence structure.
- Gender, number, case, tense, voice, sentence correction and common usage.
- Sandhi, Samasa, prefixes/suffixes or their language-specific equivalents where applicable.
- Vocabulary: synonyms, antonyms, homonyms, idioms, proverbs and contextual meanings.
- Spelling, error correction, sentence transformation, fill-in-the-blanks and language usage.
- Basic literature awareness and prescribed school-level prose/poetry where applicable.
- School-level Urdu prose and poetry; literary forms, comprehension and interpretation.
- Urdu prose, poetry, ghazal, nazm and major literary forms; literary history and prescribed texts.

Bengali

- Reading comprehension, Bengali grammar, vocabulary, sentence structure and error correction.
- Sandhi, samasa, word formation, synonyms, antonyms, idioms, proverbs and usage.
- Prose, poetry, fiction, drama and prescribed Bengali literature; literary devices and interpretation.

Maithili

- Grammar, word formation, sentence structure, case, tense, vocabulary, idioms and proverbs.
- Comprehension of prose and poetry; meaning, inference, explanation and language usage.
- Maithili literary traditions, folk literature, poetry, prose, drama and prescribed texts.

Sanskrit

- Grammar: noun and verb forms, cases, sandhi, samasa, prefixes, suffixes, tense/mood and sentence construction.
- Translation, comprehension, unseen prose/poetry, word meanings and question-answer.
- Prescribed prose, poetry, stories, verses and basic Sanskrit literary awareness.

- Classical Sanskrit literature, major authors/works, poetic forms, figures of speech and literary interpretation.

Arabic

- Arabic reading comprehension, vocabulary and sentence structure.
- Noun/verb patterns, pronouns, gender, number, cases, tense/mood, syntax and translation.
- Prescribed prose/poetry, literary forms and basic Arabic literary history.

Persian

- Persian reading comprehension, vocabulary and grammar.
- Nouns, pronouns, verbs, tense, sentence construction, idioms and translation.
- Prescribed prose/poetry and basic Persian literary history and forms.

Magahi

- Grammar, phonology, vocabulary, word formation and sentence structure.
- Comprehension of prose/poetry, idioms, proverbs and language usage.
- Magahi folk literature, oral traditions, poetry, prose and major literary forms.

Bhojpuri

- Grammar, vocabulary, sentence structure, idioms and proverbs.
- Prose and poetry comprehension, interpretation and language usage.
- Bhojpuri folk literature, songs, stories, drama and literary traditions.

Pali

- Pali grammar, noun/verb forms, cases, sandhi, samasa and sentence construction.
- Translation and comprehension of Pali prose and verses.
- Buddhist literature, Tripitaka tradition, major texts, authors and literary history.

Prakrit

- Prakrit languages, grammar, morphology, word formation and sentence structure.
- Translation and comprehension of Prakrit prose/poetry.
- Jain literary traditions, major texts, linguistic development and relation with Sanskrit/Apabhramsha.

Mathematics

- Sets, relations and functions; real numbers; complex numbers; sequences and series; permutations and combinations; binomial theorem.
- Algebra: matrices, determinants, systems of equations, polynomials, inequalities and mathematical reasoning.
- Coordinate geometry: straight lines, circles, conic sections and three-dimensional geometry.
- Trigonometry: functions, identities, equations and inverse trigonometric functions.
- Calculus: limits, continuity, differentiation, applications of derivatives, integration and applications, differential equations.
- Vectors and three-dimensional geometry.
- Statistics and probability: distributions, measures, conditional probability and standard senior-secondary concepts.
- Problem solving, proofs, graphs and numerical applications.

Physics

- Units and measurement; vectors; kinematics; laws of motion; work, energy and power; circular/rotational motion; gravitation.
- Properties of matter, fluids, thermal physics, thermodynamics, kinetic theory, oscillations and waves.
- Electrostatics, electric potential, capacitance, current electricity and circuits.
- Magnetism, moving charges, electromagnetic induction, alternating current and electromagnetic waves.
- Ray optics, wave optics, optical instruments and interference/diffraction basics.
- Dual nature of matter/radiation, atoms, nuclei, radioactivity and semiconductor electronics.
- Experimental physics: measurements, errors, graphs, instruments and practical applications.

Chemistry

- Basic concepts, mole concept, atomic structure, periodicity, chemical bonding and molecular structure.
- States of matter, thermodynamics, equilibrium, ionic equilibrium, redox reactions and electrochemistry.
- Solutions, chemical kinetics, surface chemistry and metallurgy.
- Periodic chemistry: s/p/d/f block elements, coordination compounds and important inorganic reactions.
- Organic chemistry: hydrocarbons, halo compounds, alcohols, phenols, ethers, aldehydes, ketones, carboxylic acids and amines.
- Biomolecules, polymers, chemistry in everyday life and environmental chemistry.
- Practical chemistry: qualitative analysis, titration concepts, laboratory safety and common preparations.

Botany

- Plant diversity: algae, bryophytes, pteridophytes, gymnosperms and angiosperms.
- Plant morphology and anatomy; tissues, vascular systems and adaptations.
- Cell biology, biomolecules, enzymes and cellular processes.
- Plant physiology: transport, mineral nutrition, photosynthesis, respiration, plant growth and hormones.
- Plant reproduction, genetics, molecular biology and biotechnology.
- Plant breeding, economic botany, ecology, biodiversity and conservation.
- Practical botany: specimen identification, microscopy, experiments and field-based observations.

Zoology

- Animal diversity and classification; structural organisation, tissues and basic anatomy.
- Cell biology, biomolecules, enzymes and genetics.
- Human physiology: digestion, respiration, circulation, excretion, movement, neural control and endocrine regulation.
- Reproduction, development, heredity, evolution and animal behaviour.
- Ecology, biodiversity, conservation and environmental biology.
- Applied zoology, biotechnology, health, disease and parasitology basics.
- Practical zoology: specimen identification, diagrams, experiments and observation.

History

- Ancient India: sources, prehistory, Indus civilisation, Vedic age, Mahajanapadas, Buddhism/Jainism, Mauryas, post-Mauryan period, Gupta age and early medieval India.
- Medieval India: Delhi Sultanate, regional kingdoms, Vijayanagara/Bahmani states, Mughal Empire, administration, economy, society, art and culture.
- Bhakti and Sufi traditions; regional cultures and social change.
- Modern India: European expansion, British conquest, land revenue systems, economic impact, 1857, reform movements, nationalism and freedom struggle.
- Gandhian movements: Non-Cooperation, Civil Disobedience, Quit India; revolutionary and constitutional developments.
- Constitutional reforms, transfer of power and independence.
- World history: Renaissance, Reformation, scientific/industrial revolutions, nationalism, imperialism, world wars, Russian Revolution, fascism/Nazism and decolonisation.

Geography

- Physical geography: earth structure, geomorphic processes, rocks, landforms, atmosphere, climate, oceans and hydrology.
- Climatology: pressure belts, winds, monsoon, cyclones, rainfall and climate classification.
- India: physiography, drainage, climate, soils, vegetation, agriculture, minerals, energy resources and industries.
- Population and human geography: distribution, density, migration, settlements, urbanisation and demographic processes.
- Economic geography: agriculture, industries, transport, trade, resources and regional development.
- Environment: ecosystems, hazards, climate change, conservation and sustainable development.
- Map work, topographical interpretation, GIS and remote sensing basics.

Political Science

- Political theory: liberty, equality, justice, rights, citizenship, nationalism, secularism and democracy.
- Indian Constitution: making, philosophy, Fundamental Rights, Fundamental Duties, DPSP, Parliament, executive, judiciary and constitutional bodies.
- Federalism, centre-state relations, local government, elections and political parties.
- Comparative politics and contemporary political systems.
- International relations: globalisation, security, international organisations, environment, human rights and major global issues.
- Political thinkers and major political ideas as prescribed.

Economics

- Microeconomics: demand, supply, consumer behaviour, production, cost, revenue, market structures and price determination.
- Macroeconomics: national income, money, banking, inflation, unemployment, fiscal policy, monetary policy and balance of payments.
- Indian economy: development, poverty, unemployment, agriculture, industry, services, infrastructure and human capital.
- Economic reforms, liberalisation, privatisation, globalisation and inclusive growth.
- Public finance, taxation, government budget and sustainable development.
- Statistics for economics: data collection, presentation, measures of central tendency/dispersion and basic interpretation.

Sociology

- Basic concepts: society, community, culture, socialisation, status, role, norms, values and institutions.
- Social institutions: family, kinship, marriage, religion, education, economy and polity.
- Indian society: caste, class, tribe, gender, diversity, inequality and social stratification.
- Rural/urban change, population, migration, industrialisation and globalisation.
- Social movements, media, education, health and contemporary social issues.
- Sociological thinkers and research methods at the prescribed level.

Psychology

- Introduction to psychology, methods, biological bases, sensation, perception and attention.
- Learning, memory, thinking, intelligence, creativity, motivation and emotion.
- Personality, psychological assessment and individual differences.
- Human development across childhood, adolescence, adulthood and ageing.
- Social behaviour, attitudes, groups, stress, coping and adjustment.
- Psychological disorders, mental health, therapeutic approaches and applications.

Philosophy

- Indian philosophy: Upanishadic thought, Buddhism, Jainism, Nyaya, Vaisheshika, Samkhya, Yoga, Mimamsa and Vedanta.
- Western philosophy: Greek philosophy, rationalism, empiricism, idealism, existentialism and major thinkers.
- Logic, reasoning, epistemology, metaphysics and philosophy of science.
- Ethics, aesthetics, philosophy of religion, social/political philosophy and contemporary issues.

Home Science

- Food and nutrition: nutrients, balanced diet, meal planning, deficiency disorders, food preservation, safety and hygiene.
- Human development: growth, development, family, adolescence, adulthood and ageing.
- Textiles and clothing: fibres, yarns, fabrics, dyeing, finishing, care, design and basic construction.
- Resource management: household finance, time/energy management, consumer education and decision-making.
- Interior/home management, community extension, communication and applied home science.

Computer Science

- Computer fundamentals: hardware, software, operating systems, data representation, number systems and Boolean logic.
- Programming: algorithms, flowcharts, variables, data types, operators, control structures, functions, strings, arrays/lists and file handling.

- Object-oriented programming concepts, classes/objects and basic software design.
- Data structures: stacks, queues, linked lists, trees and searching/sorting basics.
- Databases: relational model, keys, SQL, queries, normalisation basics and database applications.
- Computer networks: LAN/WAN, protocols, OSI/TCP-IP concepts, internet, web and network security.
- Cybersecurity, privacy, ethics, AI, machine learning concepts, cloud computing and IoT.

Business Studies

- Nature and significance of management; principles of management and business environment.
- Planning, organising, staffing, directing, leadership, motivation, communication and controlling.
- Business finance, financial markets, sources of finance, capital structure and investment basics.
- Marketing management: product, price, place, promotion, branding, consumer protection and digital marketing.
- Entrepreneurship, business ethics, social responsibility and case-based application.

Accountancy

- Accounting principles, journal, ledger, trial balance, cash book and bank reconciliation.
- Depreciation, bills of exchange, rectification of errors and financial statements.
- Partnership accounts: admission, retirement, death and dissolution.
- Company accounts: shares, debentures and issue/redemption concepts.
- Financial statement analysis, ratios, cash flow and computerised accounting basics.

Music

- Swara, shruti, raga, tala, laya and notation.
- Indian classical and folk traditions; vocal/instrumental forms and compositions.
- History of Indian music, major musicians, instruments, gharanas/traditions and terminology.
- Music appreciation, aesthetics and prescribed practical/theoretical content.
- Advanced raga/tala theory, notation, forms, music history, aesthetics, major schools/traditions and prescribed repertoire.

Fine Arts

- Elements and principles of art: line, shape, form, colour, texture, space, balance, rhythm and composition.
- Drawing and painting: sketching, still life, nature study, perspective, shading, colour theory and creative composition.
- Indian art history: prehistoric, ancient, medieval and modern art; folk and tribal traditions.
- Major schools, artists, artworks, techniques and basic art appreciation.

Dance

- Elements of dance: movement, rhythm, tala, laya, expression, gesture and stage presentation.
- Major Indian classical dance traditions, folk dances, repertoire, terminology and history.
- Abhinaya, mudras, basic theory, aesthetics and performance-related concepts.

Physical Education

- Health, fitness, wellness, nutrition and lifestyle.
- Anatomy and physiology basics; exercise, training methods, fitness components and assessment.
- Rules and skills of major games and sports; tournaments and sports organisations.
- Yoga, first aid, safety, sports psychology, motivation, measurement and evaluation.

Entrepreneurship

- Entrepreneurial opportunity, creativity, innovation, characteristics and competencies.
- Business idea, feasibility study, market research, business plan and forms of organisation.
- Resource mobilisation, finance, marketing, operations, human resources and risk management.
- Enterprise growth, social entrepreneurship, digital entrepreneurship, ethics and legal/regulatory basics.

Source basis: BPSC TRE-4.0 notification framework and current syllabus references. The notification gives the paper structure, broad subject areas and SCERT/NCERT curriculum alignment; this PDF expands those areas into practical topic/subtopic checklists. Verify any later BPSC corrigendum before the exam.